

HOUSE PRICE PREDICTION USING MACHINE LEARNING

Project Documentation

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Domain	Data Science
Dataset	Ames Housing Dataset
Dataset Source	Kaggle
Target Variable	SalePrice

ABSTRACT

This project focuses on predicting house prices using machine learning techniques. The Ames Housing dataset was used to study the relationship between property characteristics and sale prices. Three regression models—Linear Regression, Decision Tree Regression, and Random Forest Regression—were trained and compared using standard regression evaluation metrics. Random Forest Regression produced the best overall performance and was integrated into a Streamlit dashboard for interactive house-price prediction and analysis.

PROJECT OBJECTIVE

The main objective is to develop a model that can estimate the sale price of a house from its characteristics and to present the results through an interactive dashboard that is easy to use and understand.

KEY RESULT

Random Forest Regression

$R^2 = 0.9231$

MAE = \$15,260.49

RMSE = \$24,825.36

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1. PROJECT OVERVIEW AND DATASET

The Ames Housing dataset contains residential property information such as quality, living area, lot size, year built, garage details, bathrooms, neighborhood and other construction-related attributes. The target variable used for prediction is **SalePrice**. The dataset used in this project contains 2,930 properties.

Dataset summary

Properties	Features	Median Sale Price	Average Sale Price
2,930	81	\$160,000	\$180,796

2. METHODOLOGY

The project follows a standard machine-learning workflow: data preparation, feature engineering, model training, model evaluation and deployment through a Streamlit dashboard.

Feature engineering

Feature	Definition
TotalSF	Total Bsmt SF + 1st Flr SF + 2nd Flr SF
TotalBathrooms	Full Bath + $0.5 \times$ Half Bath + Bsmt Full Bath + $0.5 \times$ Bsmt Half Bath
TotalPorchSF	Open Porch SF + Enclosed Porch + 3Ssn Porch + Screen Porch + Wood Deck SF

Models used

Model	Description
Linear Regression	Used as a basic regression model for comparison.
Decision Tree Regression	Captures non-linear relationships through decision rules.
Random Forest Regression	An ensemble of decision trees used as the final model.

The trained Random Forest model is saved as **house_price_model.pkl** and is loaded directly by the Streamlit application. The application does not retrain the model during prediction.

3. MODEL EVALUATION

The three regression models were evaluated using R^2 , Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The recorded project results are shown below.

Model	R^2	MAE	RMSE
Linear Regression	0.887962	\$17,364.58	\$29,971.12
Decision Tree Regression	0.814346	\$25,049.93	\$38,580.96
Random Forest Regression	0.923131	\$15,260.49	\$24,825.36

Result discussion

Random Forest Regression achieved the highest R^2 score among the three models. It also produced the lowest recorded MAE and RMSE, indicating that its predictions had smaller errors on the test set.

An R^2 score of 0.9231 means that the model explains approximately 92.31% of the variance in house prices in the test set. MAE represents the average absolute prediction error, while RMSE measures the typical error magnitude while giving greater weight to larger errors.

The exact MSE value is not included in this report because it was not available in the final supplied results. No value has been estimated from the rounded RMSE.

4. DASHBOARD

The Streamlit dashboard provides an interface for entering property details and obtaining a price prediction from the trained Random Forest model. It also contains market-analysis graphs and model-diagnostic visualizations.

4. DASHBOARD — PROPERTY VALUATION

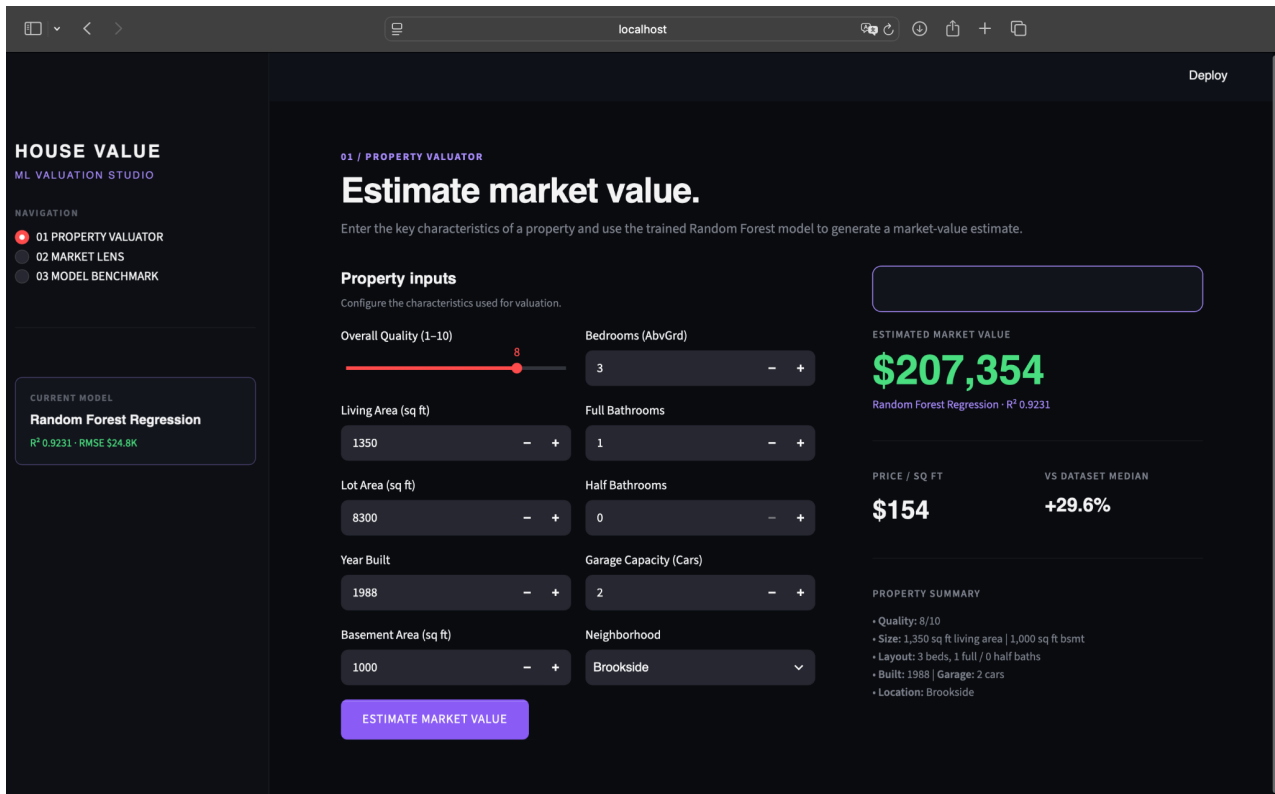


Figure 1. Property Valuator interface showing property inputs and the predicted market value.

The user can enter features including overall quality, living area, lot area, year built, bedrooms, bathrooms, garage capacity, basement area and neighborhood. After selecting the prediction button, the application displays the estimated house price and supporting information.

5. MARKET ANALYSIS

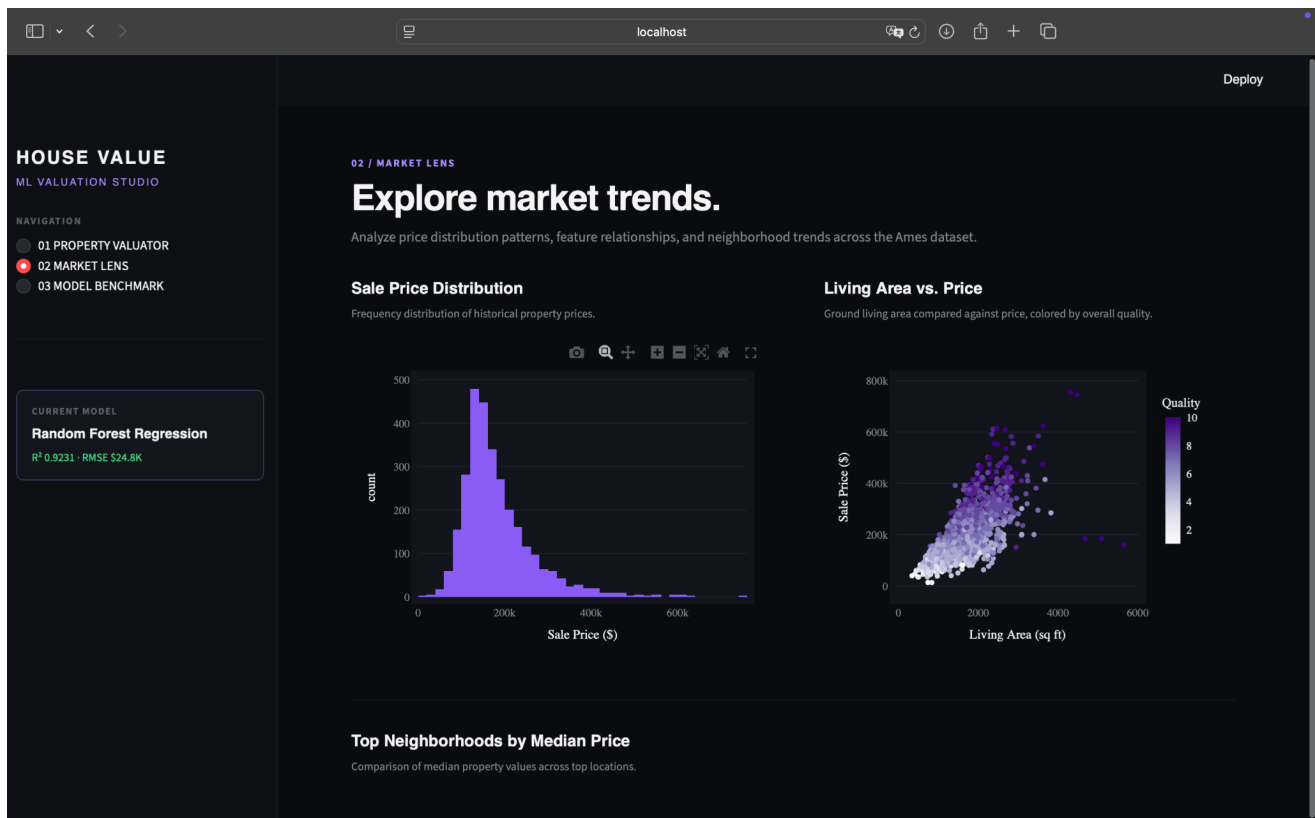


Figure 2. Market Lens showing the distribution of house prices and the relationship between living area and sale price.

The dashboard uses exploratory graphs to understand the dataset before interpreting the model. The Sale Price Distribution shows the range and concentration of property prices. The Living Area vs. Sale Price graph helps illustrate how house size is related to price.

Additional dashboard analysis includes average sale price by neighborhood and the strongest numerical correlations with SalePrice. Correlation is used here for exploratory analysis and is separate from Random Forest feature importance.

6. MODEL DIAGNOSTICS AND CONCLUSION

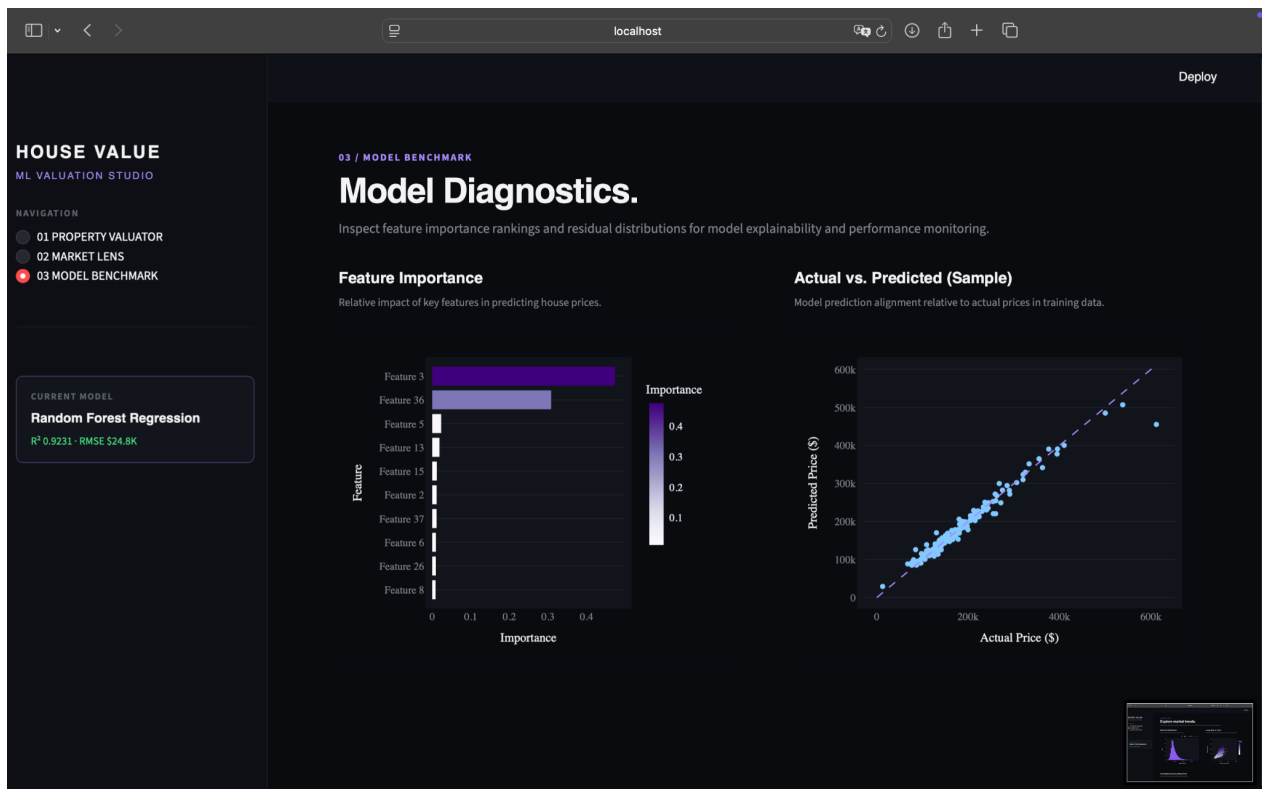


Figure 3. Model diagnostics showing Random Forest feature importance and Actual vs. Predicted prices.

Feature importance is used to identify the variables that contribute most to the Random Forest predictions. The Actual vs. Predicted graph provides a visual comparison between the model's predictions and the observed sale prices.

Conclusion

This project developed a house-price prediction system using the Ames Housing dataset and three regression algorithms. Random Forest Regression performed best, achieving an R^2 score of 0.9231, an MAE of approximately \$15,260 and an RMSE of approximately \$24,825.

The trained model was integrated into a Streamlit dashboard so that users can enter property characteristics and obtain a prediction while also exploring market patterns and model results. The project demonstrates the use of Data Science techniques from data preparation and feature engineering through model evaluation and application development.

References

1. Ames Housing Dataset — Kaggle.
2. House Price Prediction Jupyter Notebook.
3. Streamlit dashboard source code (app.py).